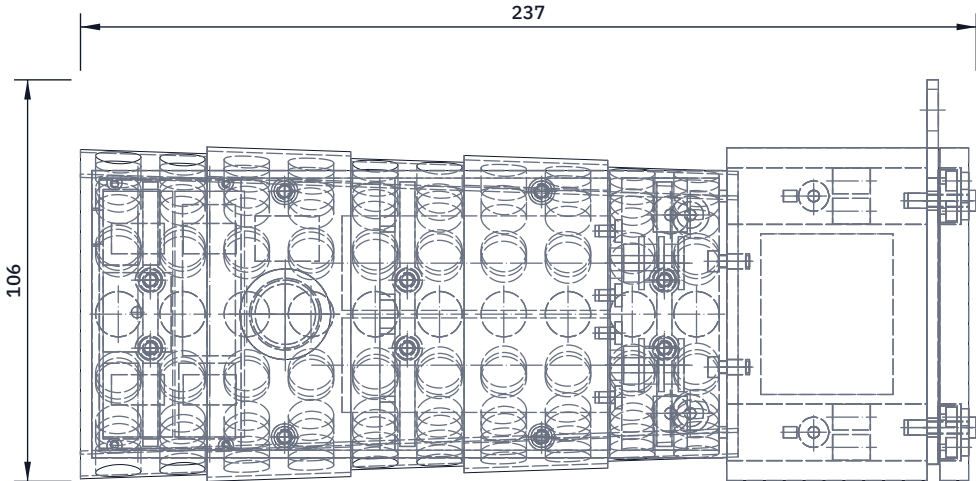


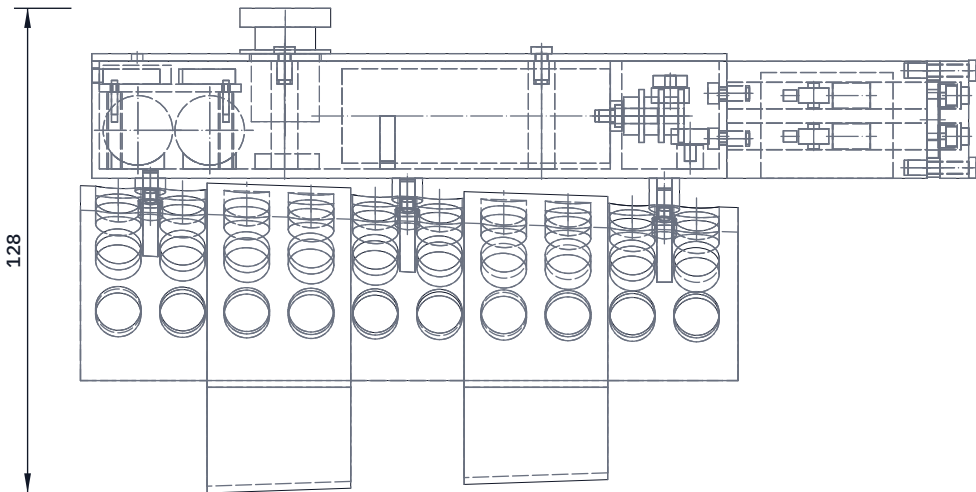
CONCEPT, NOT FOR FABRICATION

FlexHand · github.com/BoujeeEnjinia1701/flexhand



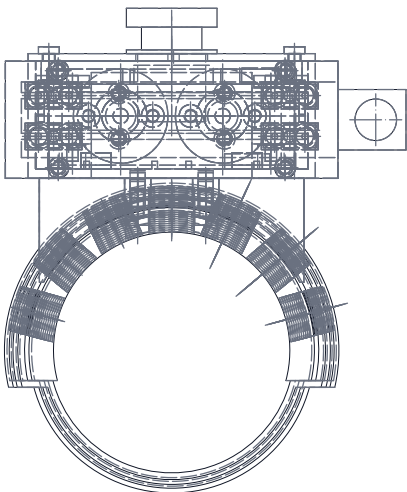
TOP VIEW

Scale 1:2; looking down (along -Z)



FRONT VIEW

Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)

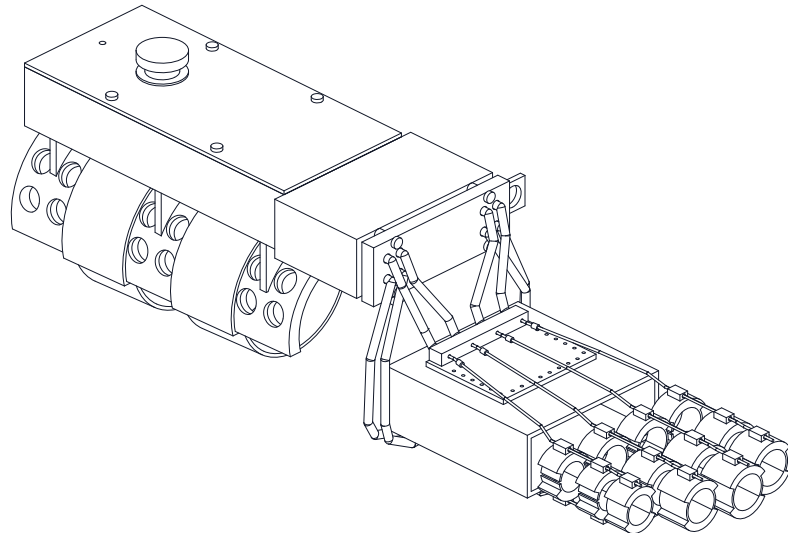
PARTS LIST (ITEMS MATCH BOM/BOM.CSV)

- 1 Forearm cuff with liner and straps
- 2 Pack base
- 3 Gearmotor with encoder (2)
- 4 Antagonistic spool (2)
- 5 Li-ion cells and 2S BMS
- 6 Controller
- 7 Motor driver (2)
- 8 Pack lid
- 9 Emergency stop
- 10 Anchor block, release plate, cover

PARTS LIST, CONTINUED

- 11 Bowden sheaths (8)
- 12 to 17, 21 Hand side (iso only)
- 18 USB-C charger
- 19 Wiring (not shown)
- 20 Idler bearings (4)
- 22 Balance pulleys (4)
- 23 Tray; 24 Fixings; 25 5 V regulator
- Not a medical device.
- Research use under supervision.
- Never charge while worn.

REV	DESCRIPTION	DATE	BY
P1	General arrangement for TRL 3 (FXH-CAL-001)	2026-09-25	AC
P2	FXH-DDR-002: balance pulleys, 44 mm anchor block, perforated cuff, thimbles	2026-09-25	AC
P3	FXH-DDR-003: constructable design	2026-10-01	AC



ISOMETRIC VIEW, WHOLE DEVICE

Right-hand device as modelled; a left-hand device is its mirror image; not to scale; seen from the front right and above, about 30 deg elevation

MAIN DIMENSIONS (MM)

- Pack 168 long x 76 wide; walls 2, lid 2
- Pack top 78.5 above forearm axis; wrist to pack front 71
- Gearmotors 25 dia x 71, axes along forearm at $y = \pm 13.5$
- Spools: core 10 dia, flanges 14 dia, 12 wide
- Tendon exits at $y = \pm 31.25$; idlers 10 dia (623ZZ class) on 3 mm pins
- Cells 2 x 18650 across the pack at $x = -227, -208$
- Forearm cuff 174 long; shell 2, liner 4; 12 dia holes; straps 38 wide
- Anchor block 53 long x 88 wide, 64 with cover; pulleys 8 dia, 30 travel
- Anchor cuffs (index, middle, ring, little): 16.4, 18.6, 17.0, 12.2
- Thimbles (index, middle, ring, little): 17.0, 19.0, 17.5, 13.2
- Guide cuffs 12 wide; plates TPU 2.5 thick

FLEXHAND

github.com/BoujeeEnjinia1701/flexhand

Forearm unit general arrangement

DRAWING NO. FXH-DWG-001		REV P3	SHEET 1 of 1	SIZE ANSI B
SCALE 1:2	UNITS mm	PROJECTION 	DATE 2026-10-01	DRAWN Amish Chadha
MATERIAL / NOTES Pack and cuff PETG; plates and cuffs TPU 95A; bought parts per bom/bom.csv. PRELIMINARY, NOT FOR FABRICATION				
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